DC Circuits — Reference

Textbook: Ulaby, Maharbiz, Furse, *Circuit Analysis and Design*, 3rd ed., chapters 1–2. Gray tags like [book 2-1] point to the book section. The book writes voltage as υ (upsilon). These notes write v.

prefixes

Symbol	Prefix	Power	Multiply by	In words
E	exa	10^{18}		quintillion
P	peta	10^{15}		quadrillion
T	tera	10^{12}	1,000,000,000,000	trillion
G	giga	10^{9}	1,000,000,000	billion
M	mega	10^{6}	1,000,000	million
k	kilo	10^{3}	1,000	thousand
—	(none)	10^{0}	1	one
m	milli	10^{-3}	0.001	thousandth
μ	micro	10^{-6}	0.000 001	millionth
n	nano	10^{-9}	0.000 000 001	billionth
p	pico	10^{-12}	0.000 000 000 001	trillionth
f	femto	10^{-15}		quadrillionth
a	atto	10^{-18}		quintillionth

Capital M (mega) and small m (milli) are 10^{9} apart. [book Table 1-2]

quantities and units

Quantity	Symbol	Unit	In base units
charge	q, Q	coulomb (C)	A·s
current	i, I	ampere (A)	C/s
voltage	v, V (book: υ)	volt (V)	J/C
power	p, P	watt (W)	J/s = V·A
energy	w, W	joule (J)	W·s
resistance	R	ohm (Ω)	V/A
conductance	G	siemens (S)	A/V $= 1/\Omega$
resistivity	ρ (rho)	ohm-meter	Ω·m
conductivity	σ (sigma)	siemens/meter	S/m
capacitance	C	farad (F)	C/V
inductance	L	henry (H)	V·s/A
frequency	f	hertz (Hz)	1/s
length, area	ℓ, A	m, m^2	—
time	t	second (s)	—
temperature	T	kelvin (K), °C	K =°C+273

Lowercase (q, i, v, p, w) can change with time. Uppercase (Q, I, V, P, W) is a constant (dc). [book Table 1-1]

constants and conversions

Item	Value
charge of one electron	$-e$, $e = 1.6 \times 10^{-19}$ C (more exact: 1.602)
charge of one proton	$+e$
electrons in 1 C	6.25×10^{18}
minute, hour	1 min = 60 s, 1 h = 3600 s
ampere-hour	1 Ah = 3600 C, 1 mAh = 3.6 C
watt-hour, kilowatt-hour	1 Wh = 3600 J, 1 kWh $= 3.6 \times 10^{6}$ J
length	1 mm $= 10^{-3}$ m, 1 cm $= 10^{-2}$ m
area	1 $mm^2 = 10^{-6}$ m^2, 1 $cm^2 = 10^{-4}$ m^2
specific heat of water	$c = 4186$ J/(kg·°C)
mass of water	1 L ≈ 1 kg, 1 fl oz ≈ 0.02957 kg

The book uses $e = 1.6 \times 10^{-19}$ C. Use it to match the book's answers.

circuit terminology

Term	Meaning [book Table 1-4]
node	all the wire between elements: one voltage. Points joined by a wire or a short are the same node
ordinary node	connects to 2 elements
extraordinary node	connects to 3 or more elements
branch	one element between two nodes
path	a chain of branches that repeats no node
extraordinary path	a path between two adjacent extraordinary nodes
loop	a closed path
independent loop	a loop with at least one branch that no other chosen loop has
mesh	a loop that encloses no other loop
in series	same current: only ordinary nodes between them
in parallel	same voltage: they share both end nodes
planar circuit	can be drawn with no crossing wires. A crossing without a dot is not a connection
open circuit	$i = 0$, $R = \infty$, any voltage
short circuit	$v = 0$, $R = 0$, any current
ground	the 0 V reference node. Chassis ground: the metal frame. Earth ground: a rod in the soil

conductivity and resistivity, 20°C

Material	σ (S/m)	$\rho = 1/\sigma$ (Ω·m)
Conductors		
silver	6.17×10^{7}	1.62×10^{-8}
copper	5.81×10^{7}	1.72×10^{-8}
gold	4.10×10^{7}	2.44×10^{-8}
aluminum	3.82×10^{7}	2.62×10^{-8}
iron	1.03×10^{7}	9.71×10^{-8}
mercury (liquid)	1.04×10^{6}	9.58×10^{-7}
Semiconductors		
carbon (graphite)	7.14×10^{4}	1.40×10^{-5}
pure germanium	2.13	0.47
pure silicon	4.35×10^{-4}	2.30×10^{3}
Insulators		
paper	$\sim 10^{-10}$	$\sim 10^{10}$
glass	$\sim 10^{-12}$	$\sim 10^{12}$
teflon	$\sim 3.3 \times 10^{-13}$	$\sim 3 \times 10^{12}$
porcelain	$\sim 10^{-14}$	$\sim 10^{14}$
mica	$\sim 10^{-15}$	$\sim 10^{15}$
polystyrene	$\sim 10^{-16}$	$\sim 10^{16}$
fused quartz	$\sim 10^{-17}$	$\sim 10^{17}$
Common materials		
distilled water	5.5×10^{-6}	1.8×10^{5}
drinking water	$\sim 5 \times 10^{-3}$	~ 200
sea water	4.8	0.2
rubber	10^{-13}	10^{13}
Biological tissues		
blood, muscle	~ 1.5	~ 0.67
fat	~ 0.1	10

Small ρ (big σ) = good conductor. [book Table 2-1] If a problem gives ρ or σ, use the problem's value.

wire size (AWG) and cross-section area

AWG	Diameter d (mm)	Area $A = \pi d^2/4$ (m^2)
0	8.3	5.41×10^{-5}
2	6.5	3.32×10^{-5}
4	5.2	2.12×10^{-5}
6	4.1	1.32×10^{-5}
10	2.6	5.31×10^{-6}
14	1.6	2.01×10^{-6}
18	1.0	7.85×10^{-7}
20	0.8	5.03×10^{-7}

Bigger AWG number = thinner wire. [book Table 2-2]

Formula	In words
$A = \pi r^2 = \dfrac{\pi d^2}{4}$	round wire: (area) $= \pi$(radius)^2
$d = \sqrt{\dfrac{4A}{\pi}}$	(diameter) from the area
$A = w \cdot h$	rectangle: (width)(thickness), the face the current passes *through*
$A = \pi(b^2 - a^2)$	hollow tube: outer radius b, inner radius a

Convert mm to m before you square: $(1\text{ mm})^2 = 10^{-6}\text{ m}^2$. ℓ is along the current, A is across it. For a block, the current direction decides which side is ℓ.

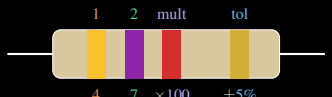
resistor color code

	Color	Digit	Multiplier	Tolerance	Temp. coef.
	black	0	$\times 1$	—	—
	brown	1	$\times 10$	$\pm 1\%$	100 ppm
	red	2	$\times 100$	$\pm 2\%$	50 ppm
	orange	3	$\times 1$k	—	15 ppm
	yellow	4	$\times 10$k	—	25 ppm
	green	5	$\times 100$k	$\pm 0.5\%$	—
	blue	6	$\times 1$M	$\pm 0.25\%$	10 ppm
	purple (violet)	7	$\times 10$M	$\pm 0.1\%$	5 ppm
	gray	8	—	—	—
	white	9	—	—	—
	gold	—	$\times 0.1$	$\pm 5\%$	—
	silver	—	$\times 0.01$	$\pm 10\%$	—

[book Fig. 2-2] Hold it so the wider gap is before the last band(s), on the right. Read left to right.

Formula	In words
4 bands: $R = (b_1 b_2) \times 10^{b_4} \pm b_5$	(two digits) $\times$ (multiplier) $\pm$ (tolerance)
5 bands: $R = (b_1 b_2 b_3) \times 10^{b_4} \pm b_5$	(three digits) $\times$ (multiplier) $\pm$ (tolerance)
6 bands: add b_6	5-band reading plus the temperature coefficient, ppm/°C

The multiplier band is the number of zeros. Digits 2–7 follow the rainbow: red, orange, yellow, green, blue, purple.



Worked: read the resistor above

1. PIECES (WIDE GAP ON THE RIGHT, READ LEFT TO RIGHT)
yellow $\to 4$, purple $\to 7$, red $\to \times 100$, gold $\to \pm 5\%$

↓

2. TEMPLATE (4 BANDS)
$R = (\boxed{\text{digit 1}}\boxed{\text{digit 2}}) \times \boxed{\text{multiplier}} \pm \boxed{\text{tolerance}}$

↓

3. PLACE
$R = (47) \times 100 \pm 5\% = 4.7\text{ k}\Omega$; range 4465 to 4935 Ω

Quick example	Work $\to$ answer
brown, black, orange, gold	$10 \times 10^3 = 10\text{ k}\Omega, \pm 5\%$
5 bands: brown, black, black, red, brown	$100 \times 10^2 = 10\text{ k}\Omega, \pm 1\%$
red, red, gold, gold	$22 \times 0.1 = 2.2\,\Omega, \pm 5\%$
(reverse) $330\,\Omega, \pm 5\%$	33×10^1: orange, orange, brown, gold

formula sheet (symbols only)

2. charge, current $Q = It$, $i = \dfrac{dq}{dt}$, $\Delta Q = \int_{t_1}^{t_2} i\,dt$, $n = \dfrac{|Q|}{e}$

3. voltage $v_{ab} = \dfrac{dw}{dq}$, $V_{ab} = V_a - V_b$, $W = QV$

4. power, energy $p = vi$, $\sum p = 0$, $w = \int p\,dt$, $W = Pt$, $P_{av} = \dfrac{1}{T}\int_0^T p\,dt$

5. extras $\text{Wh} = V \cdot \text{Ah}$, $t = \dfrac{\text{Ah}}{I}$, $E = cm\Delta T$, $\text{kWh} = \dfrac{\text{J}}{3.6 \times 10^6}$

6. sources $V_L = \dfrac{R_L}{R_s + R_L} v_s$; dependent: αv_x, $r i_x$, $g v_x$, βi_x

7. resistors $V = IR$, $R = \dfrac{\rho \ell}{A} = \dfrac{\ell}{\sigma A}$, $P = I^2R = \dfrac{V^2}{R}$, $R = R_0(1 + \alpha T)$

8. KCL, KVL $\sum i_{\text{in}} = \sum i_{\text{out}}$, $\sum v = 0$; N unknowns: M KVL + $(N - M)$ KCL

9. combine $R_1 + R_2$, $\dfrac{R_1 R_2}{R_1 + R_2}$, $G_{eq} = \sum G$

10. dividers $v_1 = \dfrac{R_1}{R_1 + R_2} v_s$, $i_1 = \dfrac{R_2}{R_1 + R_2} i_s = \dfrac{G_1}{G_{eq}} i_s$

11. source transf. $i_s = \dfrac{v_s}{R_s}$, $v_s = i_s R_s$ (same R_s)

12. $\Delta \to Y$ $R_1 = \dfrac{R_b R_c}{R_a + R_b + R_c}$, $R_a = \dfrac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$

13. bridge $R_x = \dfrac{R_2}{R_1} R_3$, $V_{out} \approx \dfrac{V_0}{4} \dfrac{\Delta R}{R}$, $R = R_0(1 + \alpha P)$

14. diode, LED forward: $I = \dfrac{V_s - V_F}{R}$, $V_F \approx 0.7$ V (Si)

calculus and algebra tools

Derivatives (a, ω, n, c constants):

Function	Derivative
c, t^n	0, nt^{n-1}
e^{at}	$a e^{at}$
$\sin(\omega t)$, $\cos(\omega t)$	$\omega\cos(\omega t)$, $-\omega\sin(\omega t)$
$f \cdot g$ (product)	$f'g + fg'$
$f(g(t))$ (chain)	$f'(g) \cdot g'$; e.g. $\sin^2(\omega t) \to 2\omega\sin(\omega t)\cos(\omega t)$

Integrals:

Function	Antiderivative
t^n	$\dfrac{t^{n+1}}{n+1}$
e^{at}	$\dfrac{e^{at}}{a}$
$\sin(\omega t)$, $\cos(\omega t)$	$-\dfrac{\cos(\omega t)}{\omega}$, $\dfrac{\sin(\omega t)}{\omega}$
$e^{at}\cos(bt)$	$\dfrac{e^{at}(a\cos bt + b\sin bt)}{a^2 + b^2}$
$e^{at}\sin(bt)$	$\dfrac{e^{at}(a\sin bt - b\cos bt)}{a^2 + b^2}$
$\int_0^\infty Ae^{-at}\,dt$	$\dfrac{A}{a}$ $(a > 0)$

Definite integral: $F(\text{top}) - F(\text{bottom})$. $e^0 = 1$. $e^{-at} \to 0$ as $t \to \infty$.
Trig identities: $\sin x \cos x = \frac{1}{2}\sin 2x$, $\cos^2 x = \frac{1}{2}(1+\cos 2x)$, $\sin^2 x = \frac{1}{2}(1-\cos 2x)$.
Maximum or minimum: set the derivative to 0 and solve for t. Example: largest q where $\frac{dq}{dt} = 0$; largest p where $\frac{dp}{dt} = 0$.

Solving simultaneous equations

2 unknowns ($a_1x + b_1y = c_1$, $a_2x + b_2y = c_2$), Cramer's rule:

$$D = a_1b_2 - a_2b_1 \qquad x = \frac{c_1b_2 - c_2b_1}{D} \qquad y = \frac{a_1c_2 - a_2c_1}{D}$$

3 unknowns:

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2).$$

$x_k = \frac{D_k}{D}$, where D_k is D with column k replaced by the right-hand sides. Or: eliminate one unknown at a time, or use a calculator's matrix solver (rref).
[book App. B]

Quick example	Work → answer
$3x+2y=12$, $x-y=-1$	$D = 3(-1)-1(2) = -5$; $x = \frac{12(-1)-(-1)(2)}{-5} = 2$, $y = \frac{3(-1)-1(12)}{-5} = 3$

which tool? — given → find

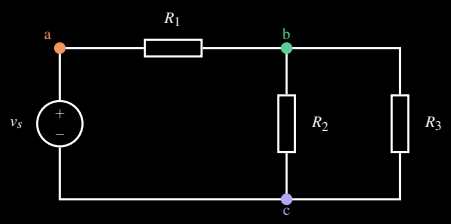
The number in parentheses is the concept section.

Given → find	Do this	
a circuit → nodes, series, parallel	terminology table	(1)
charge Q → electrons	$n = \lvert Q\rvert/e$	(2)
$q(t) \to i$; $i(t) \to \Delta Q$	differentiate; integrate	(2)
a $q(t)$ or $i(t)$ graph	slope; area	(2)
node voltages, new ground	subtract the new ground's old value	(3)
V, I → power, sign	$P = VI$, passive sign convention	(4)
$v(t)$, $i(t)$ → energy, average power	$\int vi\,dt$; $\frac{1}{T}\int p\,dt$	(4)
battery, kWh, heat, cost	extras	(5)
switch at t_0	redraw: closed = short, open = open	(6)
two of V, I, R → the third	Ohm's law	(7)
wire: $R \leftrightarrow \ell$, A, material	$R = \rho\ell/A$ and the tables	(7)
resistance at a new temperature	$R = R_0(1+\alpha T)$	(7)
color bands → R	color chart (ref)	
currents at a node; voltages in a loop	KCL; KVL	(8)
N unknown currents	M KVL + $(N-M)$ KCL, solve	(8)
network → R_{eq}	combination chart	(9)
series → v share; parallel → i share	dividers	(10)
mixed sources and resistors	source transformation	(11)
not series, not parallel (a bridge)	$\Delta \leftrightarrow Y$	(12)
bridge balance; sensor output	Wheatstone	(13)
force on a sensor → V_{out}	piezoresistor, cantilever	(13)
diode or LED in a circuit	assume on, check $I > 0$	(14)

DC Circuits — Concepts

1. reading a circuit [book 1-3]

Label the nodes first. Every wire segment between elements is one node, even if it is drawn long. Merge points joined by a wire or a short.
Puzzle pieces. In every diagram example, a colored piece in the picture goes in the slot [of that color]. Order: **pieces** → **template** → **place** → **solve**.



Worked: classify the circuit above

1. PIECES (WHAT TOUCHES EACH NODE)
a: v_s, R_1 (2) b: R_1, R_2, R_3 (3) c: v_s, R_2, R_3 (3)

↓

2. TEMPLATES
series: only an [ordinary node (2 elements)] between them
parallel: they share [one end node] and [the other]

↓

3. PLACE
a ordinary ⇒ v_s, R_1 series. R_2, R_3 share b and c ⇒ parallel.
Meshes: (v_s, R_1, R_2) and (R_2, R_3) = 2.

Meters [book Fig. 1-18]: a voltmeter goes in parallel (across the element), red lead on +; ideal: no current. An ammeter goes in series (in the path); ideal: no drop.
Same branch, same direction = same current. Opposite arrows in one branch are negatives.

2. charge and current [book 1-4]

Current is the flow of positive charge. Electrons ($-e$) moving one way count as current the other way. Protons ($+e$) count in their own direction.

Formula	In words
$n = \dfrac{\lvert Q\rvert}{e}$, $Q = ne$	(electrons) = (charge) ÷ (charge of one)
$Q = It$, $I = \dfrac{Q}{t}$, $t = \dfrac{Q}{I}$	(charge) = (current)(time), constant current
$i(t) = \dfrac{dq}{dt}$	(current) = rate of change of (charge): differentiate
$q(t) = \int_{-\infty}^{t} i\,dt$	(charge so far) = total of (current): integrate
$\Delta Q = q(t_2) - q(t_1) = \int_{t_1}^{t_2} i\,dt$	(charge moved from t_1 to t_2)
graph of $q(t)$	current = slope of each straight piece
graph of $i(t)$	charge = area under it (piece by piece)

Quick example	Work → answer
2 mC → electrons	$\dfrac{2\times10^{-3}}{1.6\times10^{-19}} = 1.25\times10^{16}$
3×10^{18} electrons/s go $+x$; 10^{18} protons/s go $-x$	both are current toward $-x$: $4\times10^{18}\cdot1.6\times10^{-19} = 0.64$ A in $-x$
90 C in 30 s; 0.5 A for 2 min	3 A; $0.5\cdot120 = 60$ C
$q = 3t^2$ mC → i	$6t$ mA
$q = 0.8(1-e^{-0.5t})$ nC → i	$0.8\cdot0.5e^{-0.5t} = 0.4e^{-0.5t}$ nA
q rises 0 → 6 C in a line over 0–3 s	slope: $6/3 = 2$ A
$i = 4$ A for 2 s, then 0	area: $4\cdot2 = 8$ C
$q = 10te^{-t}$ C, when is q largest?	$\frac{dq}{dt} = 10e^{-t}(1-t) = 0 \Rightarrow t = 1$ s, $q = 10/e = 3.68$ C
$i = 0.4e^{-0.5t}$ nA, total to ∞	$\frac{A}{a} = \frac{0.4}{0.5} = 0.8$ nC

Time in seconds. Given charge, find current: differentiate. If your answer has a higher power of t, you integrated.

Worked: (integrate) $i(t) = 20e^{-0.5t}$ mA from $t = 0$ to 4 s

1. ANTIDERIVATIVE
$F(t) = \dfrac{20}{-0.5}e^{-0.5t} = -40e^{-0.5t}$

↓

2. TOP ($t = 4$)
$F(4) = -40e^{-2} = -40(0.1353) = -5.413$

↓

3. BOTTOM ($t = 0$)
$F(0) = -40e^{0} = -40$ *($e^0 = 1$)*

↓

4. TOP MINUS BOTTOM
$-5.413 - (-40) = 34.59$ mC *($mA\cdot s = mC$)*

Worked: (product rule) $q(t) = 0.5t\cos(100\pi t)$ μC

1. PARTS
$f = 0.5t$, $f' = 0.5$. $g = \cos(100\pi t)$, $g' = -100\pi\sin(100\pi t)$

↓

2. $f'g + fg'$
$i = 0.5\cos(100\pi t) - 50\pi t\sin(100\pi t)$ μA

3. voltage, node voltage, and ground [book 1-5.1]

Voltage is energy per charge between two points. It is *across* an element. A **node voltage** is measured from ground (0 V).

Formula	In words
$v_{ab} = \dfrac{dw}{dq}$	(voltage of a over b) = (energy) per (charge)
$V = \dfrac{W}{Q}$, $W = QV$	constant case
$V_{ab} = V_a - V_b$	(voltage between two nodes) = (node voltage a) − (node voltage b)
$V_{ab} = -V_{ba}$	reversing the order flips the sign

Moving ground shifts every node voltage by the same amount. Voltage *differences* do not change.

Quick example	Work → answer
30 J moves 5 C	$V = 30/5 = 6$ V
$V_a = 9$, $V_b = 5$, $V_c = 0$	subtract 5: $V_a = 4$, $V_b = 0$, $V_c = -5$ V
(ground c) → ground at b	
V_{ab} before and after	$9-5 = 4$ and $4-0 = 4$: same

4. power and energy [book 1-5.2]

Formula	In words
$p = v\,i$	(power) = (voltage)(current)
$I = \dfrac{P}{V}$, $V = \dfrac{P}{I}$	rearranged
$\sum_k p_k = 0$	conservation: (power supplied) = (power received)
$w = \int_{t_1}^{t_2} p\,dt$	(energy) = total of (power) over time
$W = Pt$, $P = \dfrac{W}{t}$, $t = \dfrac{W}{P}$	constant power
$W = V\,I\,t$	(energy) = (voltage)(current)(time)
$P_{av} = \dfrac{1}{T}\int_0^T p\,dt$	(average power) = (energy over T) ÷ T

Passive sign convention: i enters the + side of v ⇒ $p = +vi$. Enters the − side ⇒ $p = -vi$. Then $p > 0$: the device receives (absorbs) power. $p < 0$: it supplies power.



Worked: passive sign: who receives, who supplies

1. PIECES
the side the arrow enters: $+$ or $-$; the values v, i

↓

2. TEMPLATE
$p = \boxed{+}/\boxed{-}\, vi$ $\quad p > 0$ receives, $\quad p < 0$ supplies

↓

3. PLACE
5 V, 2 A into $+$: $+(5)(2) = +10$ W receives into $-$: $-(5)(2) = -10$ W supplies
-3 V, 4 A into $+$: $+(-3)(4) = -12$ W supplies

Quick example	Work → answer
three devices: $+20$, -50, p_3	$20 - 50 + p_3 = 0 \Rightarrow p_3 = 30$ W
24 W for 1 h	$24 \cdot 3600 = 86{,}400$ J
$p = 10t$ W over 0–2 s → P_{av}	$\frac{1}{2}\int_0^2 10t\,dt = \frac{1}{2}(20) = 10$ W
$p(t)$ graph	energy = area under p; build $p = v \cdot i$ piece by piece
$v = 50(1 - e^{-0.5t})$ V, $i = 4e^{-0.5t}$ mA at $t = 1$	$19.67 \cdot 2.426 = 47.7$ mW

$\int vi\,dt \neq \int v\,dt \cdot \int i\,dt$. Multiply first. Maximum power: set $\frac{dp}{dt} = 0$.

Worked: total energy, $v = 50(1 - e^{-0.5t})$ V, $i = 4e^{-0.5t}$ mA, $0 \to \infty$

1. MULTIPLY
$p = 200\left(1 - e^{-0.5t}\right)e^{-0.5t}$ mW

↓

2. DISTRIBUTE, COMBINE EXPONENTS
$p = 200e^{-0.5t} - 200e^{-t}$ $\quad (e^{-0.5t}e^{-0.5t} = e^{-t})$

↓

3. ANTIDERIVATIVE OF EACH TERM
$F(t) = -400e^{-0.5t} + 200e^{-t}$

↓

4. TOP ($t \to \infty$) AND BOTTOM ($t = 0$)
Top: 0. Bottom: $-400 + 200 = -200$

↓

5. TOP MINUS BOTTOM
$0 - (-200) = 200$ mJ

5. batteries, utility energy, heat, cost

Course-assignment topics. Ah is charge; Wh and kWh are energy.

Formula	In words
$\text{Wh} = V \cdot \text{Ah}$	(energy rating) = (voltage)(charge rating)
$t = \frac{\text{Ah}}{I} = \frac{\text{Wh}}{P}$	(run time, hours)
$\text{J} = \text{Wh} \cdot 3600$, $\quad \text{C} = \text{Ah} \cdot 3600$	hours to seconds
$n = \frac{W}{W_{\text{one}}}$	(number of batteries filled), round down
$E = cm\Delta T$	(heat) = (specific heat)(mass)(temperature change)
$\Delta T = \frac{E}{cm}$, $\quad m = \frac{E}{c\Delta T}$	rearranged
$\text{cost} = \frac{\text{J}}{3.6 \times 10^6} \cdot \text{rate}$	joules → kWh → cost

Quick example	Work → answer
2.5 Ah, 500 mA load	$2.5/0.5 = 5$ h
12 V, 60 Wh, 250 mA load	$P = 3$ W; $\quad 60/3 = 20$ h
1.08 MJ fills 3000 mAh, 3.7 V batteries	one: 11.1 Wh = 39,960 J; $\quad 27.03 \to 27$
0.5 kg water, 20 → 80°C	$4186 \cdot 0.5 \cdot 60 = 125{,}580$ J
125,580 J at 15 cents/kWh	$0.03488 \cdot 15 = 0.523$ cents

6. circuit elements, sources, switches [book 1-6]

Active devices need a power source (transistors, ICs). **Passive** devices do not (resistors, capacitors, inductors).

Element	Rule
ideal voltage source	$v = V_s$ for any current. i–v graph: vertical line
ideal current source	$i = I_s$ for any voltage. i–v graph: horizontal line
real voltage source	v_s in series with R_s. Ignore R_s if $R_s \ll R_L$
real current source	i_s in parallel with R_s. Ignore R_s if $R_s \gg R_L$
SPST switch	closed = short, open = open. The label $t = t_0$ is when it changes
SPDT switch	the common terminal moves from contact 1 to contact 2 at t_0

Formula	In words
$V_L = \frac{R_L}{R_s + R_L} v_s$	(load voltage) of a real source
$v = \alpha v_x$	VCVS: α has no units
$v = r i_x$	CCVS: r in V/A = Ω
$i = g v_x$	VCCS: g in A/V = S
$i = \beta i_x$	CCCS: β has no units

Quick example	Work → answer
9 V, $R_s = 1\,\Omega$, $R_L = 8\,\Omega$	$V_L = \frac{8}{9} \cdot 9 = 8$ V
CCVS $10 i_x$, $i_x = 0.5$ A	5 V
VCCS $0.2 v_x$, $v_x = 10$ V	2 A
switch closes at $t = 2$ s	two circuits: $t < 2$ (open branch), $t > 2$ (short)

Dependent sources are models, not energy sources. Solve them like any source: their value comes from v_x or i_x.

7. Ohm's law and resistance [book 2-1]

Formula	In words
$V = IR$	(voltage) = (current)(resistance)
$I = \frac{V}{R}$, $\quad R = \frac{V}{I}$	rearranged
$I = \frac{V_a - V_b}{R}$	current from a to b through R
$G = \frac{1}{R}$, $\quad I = GV$	(conductance) = 1÷(resistance)

Ohm's law follows the passive sign convention. If I enters the $-$ side, $I = -V/R$.

Resistance of a shape:

Formula	In words
$R = \frac{\rho\ell}{A} = \frac{\ell}{\sigma A}$	(resistance) = (resistivity)(length) ÷ (area)
$\ell = \frac{RA}{\rho} = R\sigma A$	(length) = (resistance)(area) ÷ (resistivity)
$A = \frac{\rho\ell}{R}$	(area) = (resistivity)(length) ÷ (resistance)
$\rho = \frac{RA}{\ell}$, $\quad \sigma = \frac{\ell}{RA}$	identify the material from the table

Temperature [book P2.5]:

Formula	In words
$R = R_0(1 + \alpha T)$	R_0 at 0°C, T in °C, α in °C^{-1}
$T = \frac{R/R_0 - 1}{\alpha}$	(temperature) for a given R
$\frac{R_2}{R_1} = \frac{1 + \alpha T_2}{1 + \alpha T_1}$	compare two temperatures

Copper: $\alpha = 4 \times 10^{-3}$ °C^{-1}. A **thermistor** is built to change a lot with T: $\Delta R = k\Delta T$.

Power and power rating:

Formula	In words
$P = VI = I^2R = \dfrac{V^2}{R} = GV^2$	power in a resistor (always received, becomes heat)
$R = \dfrac{V^2}{P}, \quad R = \dfrac{P}{I^2}$	(resistance) from power
$I_{max} = \sqrt{\dfrac{P_{rated}}{R}}, \quad V_{max} = \sqrt{P_{rated}R}$	largest safe current, voltage
$\text{fraction} = \dfrac{P_{load}}{P_{source}} = \dfrac{I^2R_L}{VI}$	share of the power that reaches the load

Variable resistors [book Table 2-3]: a **rheostat** (2 terminals) sets 0 to R_{max}. A **potentiometer** (3 terminals) splits R_{max}: $R_{13} + R_{23} = R_{max}$.

Quick example	Work → answer
10 V across 2 kΩ	$I = 5$ mA
100 W lamp at 120 V	$R = 120^2/100 = 144\,\Omega$
2 W in 50 Ω	$I = \sqrt{2/50} = 0.2$ A
$\frac{1}{4}$ W rating, 1 kΩ	$I_{max} = \sqrt{0.25/1000} = 15.8$ mA
carbon film: $\ell = 2$ cm, 1 mm wide, 0.1 mm thick	$A = 10^{-3} \cdot 10^{-4} = 10^{-7}$ m^2; $R = \frac{1.40\times10^{-5}\cdot0.02}{10^{-7}} = 2.8\,\Omega$
twin cable 10 m long, AWG-14 copper	both wires: $\ell = 20$ m; $R = \frac{1.72\times10^{-8}\cdot20}{2.01\times10^{-6}} = 0.171\,\Omega$
12 V, cable 0.1 Ω, load 2.9 Ω	$I = 4$ A; $\frac{16\cdot2.9}{12.4} = \frac{46.4}{48} = 0.967$
copper $R_0 = 100\,\Omega$ at 50°C	$100(1 + 0.004 \cdot 50) = 120\,\Omega$
when is copper R 10% above R_0?	$T = \frac{1.1-1}{0.004} = 25°\text{C}$
10 kΩ pot, wiper 30% from terminal 1	$R_{13} = 3$ kΩ, $R_{23} = 7$ kΩ

Wire design chain	Copper, AWG-18, target $R = 0.5\,\Omega$
1. R (from V, I if given)	$R = \dfrac{V}{I}$; here 0.5 Ω is given
2. d from the AWG table	$1.0\text{ mm} = 1.0 \times 10^{-3}$ m
3. area	$A = \dfrac{\pi(1.0\times10^{-3})^2}{4} = 7.854\times10^{-7}$ m^2
4. length	$\ell = \dfrac{RA}{\rho} = \dfrac{0.5 \cdot 7.854\times10^{-7}}{1.72\times10^{-8}} = 22.8$ m

A two-wire cable is twice as long as the run. Current goes out on one wire and back on the other.

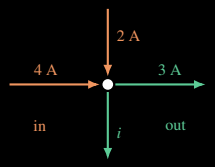
8. Kirchhoff's laws [book 2-2]

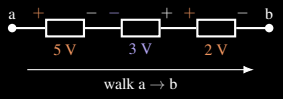
Formula	In words
$\sum i_{\text{in}} = \sum i_{\text{out}}$	KCL: (current into a node) = (current out)
$\sum_n i_n = 0$	KCL, entering = +, leaving = − (or the reverse)
$\sum_n v_n = 0$	KVL: voltages around a closed loop add to 0
$\sum v_{\text{rises}} = \sum v_{\text{drops}}$	KVL, other form
V_{ab} = sum along any path a→b	path rule: $+v$ if you meet + first

KVL sign rule: go clockwise. Enter an element at +: write $+v$. Enter at −: write $-v$. KCL also holds for a closed boundary around several nodes.

How many equations? [book 2-2.3]

1. Count the unknowns: N.
2. Write M KVL equations. Skip loops through a current source. Each new loop must use an element not used before.
3. Write $N - M$ KCL equations at extraordinary nodes.
4. Put Ohm's law in, then solve the system (Cramer's rule, reference page).





Worked: KCL at the dot (left)

1. PIECES
in: 4 A, 2 A out: 3 A, i

↓

2. TEMPLATE
$\boxed{\text{in}} + \boxed{\text{in}} = \boxed{\text{out}} + \boxed{\text{out}}$

↓

3. PLACE, SOLVE
$4 + 2 = 3 + i \Rightarrow i = 3$ A

Worked: V_{ab} by walking the path (right)

1. PIECES (SIGN MET FIRST)
+ first: 5 V, 2 V − first: 3 V

↓

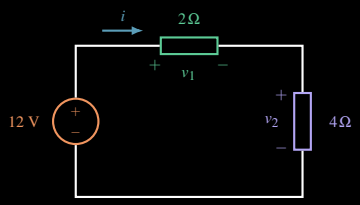
2. TEMPLATE
$V_{ab} = \boxed{+v}\ \boxed{-v} \cdots$ one term per element, in walking order

↓

3. PLACE, SOLVE
$V_{ab} = +5 - 3 + 2 = 4$ V

Quick example	Work → answer
1 A in; 3 A and i out	$i = -2$: 2 A in, against the arrow
10 A in; i_x and $1.5\,i_x$ out	$10 = 2.5\,i_x \Rightarrow i_x = 4$ A
12 V source, 5 V on R_1, find v_2	$-12 + 5 + v_2 = 0 \Rightarrow 7$ V
36 V, 2 Ω, 4 Ω, CCVS $3i$ in one loop	$36 = 2i + 4i + 3i \Rightarrow i = 4$ A
3 unknown currents, 1 current source	M = loops that avoid the source; rest KCL

A negative answer is fine. The current flows opposite to your arrow.



Worked: KVL clockwise around the loop above

1. PIECES (SIGN MET FIRST, FROM BOTTOM LEFT)
source at − v_1 at + v_2 at +

↓

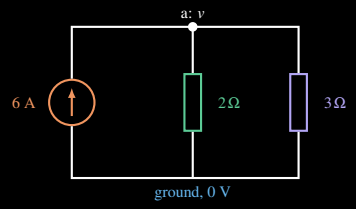
2. TEMPLATE
$\boxed{\pm\ \text{source}} + \boxed{\pm v_1} + \boxed{\pm v_2} = 0$, Ohm: $v = iR$

↓

3. PLACE, SOLVE
$-12 + 2i + 4i = 0 \Rightarrow i = 2$ A; $v_1 = 4$ V, $v_2 = 8$ V

↓

4. CHECK (POWER)
$8 + 16 - 24 = 0$ ✓ *(source supplies 24 W)*



Worked: KCL with a node voltage (above)

1. PIECES (STAND AT a, CROSS EACH R)
6 A in $2\Omega \to$ ground $3\Omega \to$ ground

↓

2. TEMPLATE
$$\boxed{\text{in}} = \frac{v - \boxed{\text{far}}}{\boxed{R}} + \frac{v - \boxed{\text{far}}}{\boxed{R}}$$

↓

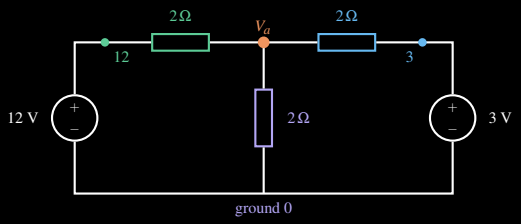
3. PLACE, SOLVE
$$6 = \frac{v-0}{2} + \frac{v-0}{3} = \frac{5v}{6} \Rightarrow v = 7.2 \text{ V}; \quad 3.6 + 2.4 = 6 \text{ A} \checkmark$$

8b. nodal analysis: the walk

Unknowns are node voltages; write one KCL per unknown node. KVL is automatic, since every drop is a difference of node voltages.

Formula	In words
$i = \dfrac{V_{\text{here}} - V_{\text{far}}}{R}$	current *leaving* through one R: stand here, cross it, note where you land
$\sum i_{\text{leaving}} = 0$	one fraction per R touching the node
source $+$ series R	far node $=$ source top $= +v_s$; $-$ on top: $-v_s$
$p_{src} = v_s \lvert i_{\text{series}} \rvert$	leaves its $+$: supplies; enters $+$: absorbs

Steps: (1) ground the node most sources touch. (2) Mark known nodes (source tops). (3) KCL at each unknown, here-node first. (4) Multiply *every term* by the LCD. (5) Solve; plug back in, each equation $= 0$.



Worked: one unknown node (above)

1. PIECES (STAND AT V_a)
left $2\Omega \to 12$ middle $2\Omega \to 0$ right $2\Omega \to 3$

↓

2. TEMPLATE
$$\frac{V_a - \boxed{\text{far}}}{\boxed{R}} + \frac{V_a - \boxed{\text{far}}}{\boxed{R}} + \frac{V_a - \boxed{\text{far}}}{\boxed{R}} = 0$$

↓

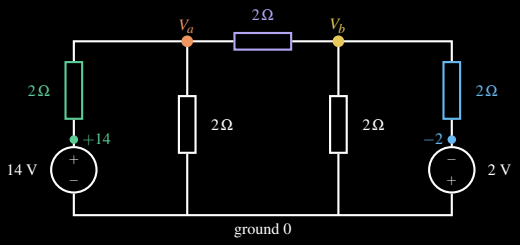
3. PLACE
$$\frac{V_a - 12}{2} + \frac{V_a - 0}{2} + \frac{V_a - 3}{2} = 0$$

↓

4. $\times 2$ EVERY TERM, SOLVE
$3V_a - 15 = 0 \Rightarrow V_a = 5$ V *(between 3 and 12)*
leaving: -3.5 A *(3.5 A in)*, 2.5 A, 1 A *(sum 0 ✓)*

↓

5. POWER: $p_{src} = \boxed{v_s} \cdot \boxed{|i_{\text{series}}|}$, $p_R = i^2 R$
12 V: 3.5 A leaves $+ \Rightarrow 12(3.5) = 42$ W supplied. 3 V: 1 A enters $+ \Rightarrow 3$ W absorbed.
resistors $24.5 + 12.5 + 2 = 39$; $39 + 3 = 42$ ✓



Worked: two unknown nodes (above)

1. PIECES
known: $+14$, -2 *(– on top)*. at V_a: $\to 14$, $\to 0$, $\to V_b$. at V_b: $\to -2$, $\to 0$, $\to V_a$

↓

2. PLACE AT V_a, $\times 2$
$$\frac{V_a - 14}{2} + \frac{V_a - 0}{2} + \frac{V_a - V_b}{2} = 0 \Rightarrow 3V_a - V_b = 14$$

↓

3. PLACE AT V_b, $\times 2$
$$\frac{V_b - (-2)}{2} + \frac{V_b - 0}{2} + \frac{V_b - V_a}{2} = 0 \Rightarrow 3V_b - V_a = -2$$

↓

4. SUBSTITUTE THE COEFFICIENT-1 VARIABLE
$V_a = 3V_b + 2 \Rightarrow 3(3V_b + 2) - V_b = 14 \Rightarrow V_b = 1$ V, $V_a = 5$ V

↓

5. CHECK
$3(5) - 1 = 14$ ✓ $3(1) - 5 = -2$ ✓

One node, one voltage: every R touching it goes in the same equation; the shared R appears in both, flipped. Minus before parentheses flips every term: $-(x-8) = -x+8$. Labeled $\pm V_x$: $+$ end minus $-$ end.

9. series and parallel resistors [book 2-3]

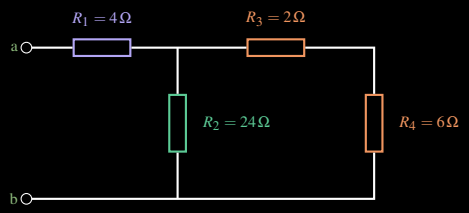
Formula	In words
$R_{eq} = R_1 + R_2 + \cdots$	**series**: resistances add
$\dfrac{1}{R_{eq}} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots$	**parallel**: reciprocals add, then flip
$R_1 \parallel R_2 = \dfrac{R_1 R_2}{R_1 + R_2}$	two in parallel: (product) ÷ (sum)
$\dfrac{R_1 R_2 R_3}{R_1 R_2 + R_2 R_3 + R_1 R_3}$	three in parallel
$G_{eq} = G_1 + G_2 + \cdots$	parallel conductances add

Series total $>$ the biggest. Parallel total $<$ the smallest. Two circuits are **equivalent** if they have the same i–v at their terminals.

Combination chart

Pattern	Formula	Example (Ω)
two in series	$R_1 + R_2$	$2 + 3 = 5$
three in series	$R_1 + R_2 + R_3$	$1 + 2 + 3 = 6$
two in parallel	$\dfrac{R_1 R_2}{R_1 + R_2}$	$6 \parallel 3 = \dfrac{18}{9} = 2$
two equal, parallel	$\dfrac{R}{2}$	$10 \parallel 10 = 5$
N equal, parallel	$\dfrac{R}{N}$	four 100s: $\dfrac{100}{4} = 25$
three in parallel (2, 3, 6)	$\dfrac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$	$\dfrac{1}{0.5 + 0.333 + 0.167} = 1$
series, then parallel	$(R_1 + R_2) \parallel R_3$	$(2+4) \parallel 3 = 6 \parallel 3 = 2$
parallel, then series	$(R_1 \parallel R_2) + R_3$	$(6 \parallel 3) + 4 = 2 + 4 = 6$
two series pairs, parallel	$(R_1 + R_2) \parallel (R_3 + R_4)$	$(1+5) \parallel (4+2) = 6 \parallel 6 = 3$
wire across (short)	$R \parallel 0 = 0$	$6 \parallel 0 = 0$
open branch in parallel	$R \parallel \infty = R$	$6 \parallel \infty = 6$
open in series	$R + \infty = \infty$	no current flows
missing R, series	$R = R_{eq} - R_1$	$1000 - 330 = 670$
missing R, parallel	$R = \dfrac{R_{eq} R_1}{R_1 - R_{eq}}$	$R \parallel 12 = 4$: $\dfrac{4 \cdot 12}{12 - 4} = 6$

How to reduce: label the nodes. Merge nodes joined by wires or shorts. Find a clear series pair or parallel pair. Replace it. Redraw. Repeat, starting farthest from the terminals.



Worked: reduce the ladder above to R_{ab}

1. PIECES (FARTHEST FROM a, b FIRST)
R_3, R_4 series R_2 parallel to that pair R_1 series with the rest

↓

2. TEMPLATES
series: $\boxed{\text{R}} + \boxed{\text{R}}$ parallel: $\dfrac{\boxed{\text{R}} \cdot \boxed{\text{R}}}{\boxed{\text{R}} + \boxed{\text{R}}}$

↓

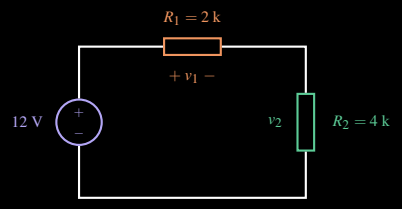
3. PLACE (INSIDE OUT)
$2+6=8$; $\dfrac{24\cdot 8}{24+8} = 6$; $R_{ab} = 4+6 = 10\,\Omega$

Worked: reduce $R_{eq} = 2 + \left[12 \parallel (4+6 \parallel 3)\right]$ Ω, inside out

INNERMOST BRACKET FIRST
$6 \parallel 3 = 2 \;\rightarrow\; 4+2=6 \;\rightarrow\; 12 \parallel 6 = 4 \;\rightarrow\; R_{eq} = 2+4 = 6\,\Omega$

10. voltage and current dividers [book 2-3]

Formula	In words
$v_1 = \dfrac{R_1}{R_1+R_2}\,v_s$	**voltage divider** (series): own R on top
$v_k = \dfrac{R_k}{R_1+\cdots+R_N}\,v_s$	N series resistors
$V_{\text{node}} = \dfrac{R_{\text{below}}}{R_T}\,v_s$	node voltage (from ground): R_{below} = sum of every R between the node and ground
$v_k = V_{\text{top}} - V_{\text{bottom}}$	across one resistor = difference of its two end nodes
$p_k = (V_{\text{top}} - V_{\text{bottom}})\,I$	series power; all p_k add to $v_s I$
$R_2 = \dfrac{R_1\,v_2}{v_s - v_2}$	resistor for a target v_2
$i_1 = \dfrac{R_2}{R_1+R_2}\,i_s = \dfrac{G_1}{G_{eq}}\,i_s$	**current divider** (parallel): other R on top, or own G
$i_k = \dfrac{R_{eq}}{R_k}\,i_s$	any number of parallel branches
$R_2 = \dfrac{R_1\,(i_s - i_2)}{i_2}$	resistor for a target i_2



Worked: voltage divider: v_1 and v_2 above

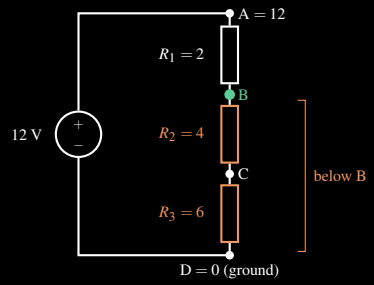
1. PIECES (FOR v_1)
own: R_1 all series: R_1, R_2 source: 12 V

↓

2. TEMPLATE
$v = \dfrac{\boxed{\text{own } R}}{\boxed{\text{every series } R}} \cdot \boxed{\text{source}}$

↓

3. PLACE
$v_1 = \dfrac{2}{2+4}\cdot 12 = 4$ V; own → R_2: $v_2 = \dfrac{4}{6}\cdot 12 = 8$ V *(4+8 = 12 ✓)*



Worked: node voltages in a series stack (above), R in Ω

1. PIECES
$R_T = 12$ between B and ground: R_2, R_3

↓

2. TEMPLATES
$V_{\text{node}} = \dfrac{\boxed{\text{sum of } R \text{ below the node}}}{\boxed{R_T}} \cdot v_s$ $v_k = \boxed{V_{\text{top}}} - \boxed{V_{\text{bottom}}}$

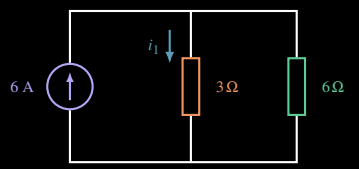
↓

3. PLACE
$V_B = \dfrac{4+6}{12}\cdot 12 = 10$ V, $V_C = \dfrac{6}{12}\cdot 12 = 6$ V, $v_2 = V_B - V_C = 4$ V *(= $\frac{4}{12}\cdot 12$ ✓)*

↓

4. POWER: $p = (V_{\text{top}} - V_{\text{bottom}})\,I$
$I = 1$ A: $p_1 = 2$, $p_2 = 4$, $p_3 = 6$ W; sum $12 = v_s I$ ✓

Node voltage = voltage across only when the other end is ground. $V_B \neq v_2$; $V_C = v_3$.



Worked: current divider: i_1 above

1. PIECES (FOR i_1)
own: 3 Ω other: 6 Ω source: 6 A

↓

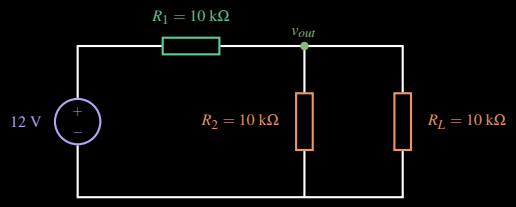
2. TEMPLATE
$i = \dfrac{\boxed{\text{other } R}}{\boxed{\text{both } R \text{ added}}} \cdot \boxed{i_s}$

↓

3. PLACE
$i_1 = \dfrac{6}{3+6}\cdot 6 = 4$ A; $i_2 = \dfrac{3}{9}\cdot 6 = 2$ A *(sum 6 ✓)*

Quick example	Work → answer
12 V, $R_1 = 9$ k, want $v_2 = 3$ V	$R_2 = \frac{9\cdot 3}{12-3} = 3$ kΩ
12 A into 2, 3, 6 Ω ($R_{eq} = 1$)	$\frac{1}{2}\cdot 12 = 6$, $\frac{1}{3}\cdot 12 = 4$, $\frac{1}{6}\cdot 12 = 2$ A
10 A, $R_1 = 4\,\Omega$, want $i_2 = 2$ A	$R_2 = \frac{4(10-2)}{2} = 16\,\Omega$

A load across a divider changes it. Combine the load with the resistor it sits across first, then divide.



Worked: the loaded divider above

1. PIECES
R_L across R_2 merge into *one* piece $R_2 \parallel R_L$ top: R_1 source: 12 V

↓

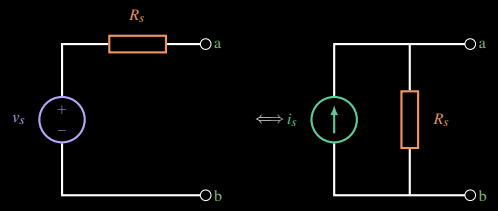
2. TEMPLATE
$$v_{out} = \frac{\boxed{R_2 \parallel R_L}}{\boxed{R_1} + \boxed{R_2 \parallel R_L}} \cdot \boxed{v_s}$$

↓

3. PLACE
$10 \parallel 10 = 5$; $v_{out} = \frac{5}{10+5} \cdot 12 = 4$ V *(unloaded: 6 V; the load pulls it down)*

11. source transformation and combining sources [book 2-3]

A voltage source v_s in series with R_s acts exactly like a current source i_s in parallel with the same R_s.



Formula	In words
$i_s = \frac{v_s}{R_s}$	voltage source → current source, same R_s in parallel
$v_s = i_s R_s$	current source → voltage source, same R_s in series
arrow of i_s	points toward the + end of v_s
$v_{eq} = v_1 - v_2 + v_3$	voltage sources in series: add with signs
$i_{eq} = i_1 - i_2 + i_3$	current sources in parallel: add with directions

Worked: transform 12 V in series with 4 Ω (left → right above)

1. PIECES
$v_s = 12$ V (+ on top) $R_s = 4\,\Omega$ in series

↓

2. TEMPLATE
$\boxed{i_s} = \frac{\boxed{v_s}}{\boxed{R_s}}$; same $\boxed{R_s}$, now parallel; arrow toward +

↓

3. PLACE
$i_s = \frac{12}{4} = 3$ A, pointing up, $\parallel 4\,\Omega$

↓

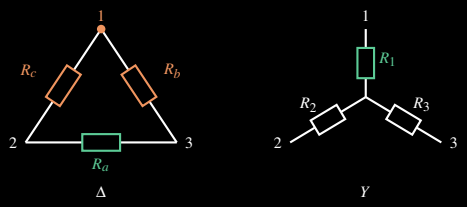
4. REVERSE (RIGHT → LEFT)
2 A $\parallel 5\,\Omega$: $v_s = 2 \cdot 5 = 10$ V in series with $5\,\Omega$

Quick example	Work → answer
9 V and 3 V in series, opposing	6 V
2 A up, 5 A up, 1 A down, in parallel	$2 + 5 - 1 = 6$ A up
open-circuit voltage of a real source	no current, no drop on R_s: $V_{oc} = v_s$

Do not transform the element whose v or i you need. It disappears into the new source.
Unrealizable: different voltage sources in parallel, or different current sources in series. Each node pair has one voltage, each branch one current. This includes dependent sources.

12. wye–delta (Y–Δ, T–Π) [book 2-4]

For networks that are neither series nor parallel (a bridge). Both shapes connect the same three nodes 1, 2, 3. The book's labels: Y arms R_1, R_2, R_3 go to nodes 1, 2, 3. Δ side R_a is opposite node 1 (between 2 and 3), R_b opposite node 2, R_c opposite node 3.



Formula	In words
$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$	Δ → Y: (Y arm) = (the two Δ sides touching its node, multiplied) ÷ (sum of all three)
$R_2 = \frac{R_a R_c}{\Sigma}$, $R_3 = \frac{R_a R_b}{\Sigma}$	same pattern
$R_a = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$	Y → Δ: (sum of pair products) ÷ (the Y arm at the node opposite)
$R_b = \frac{N}{R_2}$, $R_c = \frac{N}{R_3}$	same pattern, N = the numerator above
$R_Y = \frac{R_\Delta}{3}$, $R_\Delta = 3R_Y$	balanced (all equal)

Worked: Δ → Y with $R_a = 10$, $R_b = 20$, $R_c = 30\,\Omega$ (left figure)

1. PIECES FOR ARM R_1
sides touching node 1: $R_b = 20$, $R_c = 30$ all three: $\Sigma = 60$

↓

2. TEMPLATE
$$R_{\text{arm}} = \frac{\boxed{\text{side touching}} \cdot \boxed{\text{side touching}}}{\boxed{\text{all three added}}}$$

↓

3. PLACE
$R_1 = \frac{20 \cdot 30}{60} = 10$
node 2 touches R_a, R_c: $R_2 = \frac{10 \cdot 30}{60} = 5$
node 3 touches R_a, R_b: $R_3 = \frac{10 \cdot 20}{60} = 3.33\,\Omega$

Worked: Y → Δ, reversing the example (right figure)

1. PIECES FOR SIDE R_a
arm at the opposite node (1): $R_1 = 10$ $N = 50 + 16.7 + 33.3 = 100$

↓

2. TEMPLATE
$$R_{\text{side}} = \frac{\boxed{N}}{\boxed{\text{arm at the opposite node}}}$$

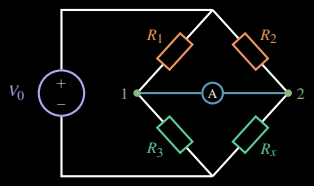
↓

3. PLACE
$R_a = \frac{100}{10} = 10$, $R_b = \frac{100}{5} = 20$, $R_c = \frac{100}{3.33} = 30\,\Omega$ *(matches the start ✓)*

Quick example	Work → answer
balanced Δ, each $30\,\Omega$	$R_Y = 10\,\Omega$

Other books swap the labels. Always match by *position*: a Y arm uses the two Δ sides that touch its node.

13. Wheatstone bridge and resistive sensors [book 2-5, 2-6.3]



Formula	In words
$V_1 = \dfrac{R_3}{R_1+R_3}V_0$, $V_2 = \dfrac{R_x}{R_2+R_x}V_0$	each side is a voltage divider
$R_x = \dfrac{R_2}{R_1}R_3$	**balanced** ($I_a = 0$, $V_1 = V_2$): (unknown) from the other three
$V_{out} = \dfrac{V_0\,\Delta R}{4R+2\Delta R}$	sensor bridge: three arms R, one $R+\Delta R$ (exact)
$V_{out} \approx \dfrac{V_0}{4}\cdot\dfrac{\Delta R}{R}$	when $\Delta R \ll R$
$\dfrac{\Delta R}{R} = \dfrac{4V_{out}}{V_0}$	(relative change) from the reading

Worked: balance: $R_1 = 100$, $R_2 = 200$, $R_3 = 150\,\Omega$, ***find*** R_x

1. PIECES
tops: 100 (left), 200 (right) bottoms: 150 (left), R_x (right)

↓

2. TEMPLATE (BALANCED: SAME RATIO EACH SIDE)
$$\frac{\boxed{\text{left top}}}{\boxed{\text{left bottom}}} = \frac{\boxed{\text{right top}}}{\boxed{\text{right bottom}}}$$

↓

3. PLACE
$$\frac{100}{150} = \frac{200}{R_x} \Rightarrow R_x = \frac{200\cdot 150}{100} = 300\,\Omega$$

Worked: sensor bridge: $V_0 = 5$ ***V,*** $\Delta R/R = 0.01$

1. PIECES
source 5 V the changed arm: $\Delta R/R = 0.01$

↓

2. TEMPLATE
$$V_{out} \approx \frac{\boxed{V_0}}{4}\cdot\boxed{\Delta R/R}$$

↓

3. PLACE
$$\frac{5}{4}\cdot 0.01 = 12.5\text{ mV} \quad \text{(exact: } 12.4\text{ mV)}$$

Piezoresistor (pressure sensor):

Formula	In words
$R = R_0(1+\alpha P)$, $\Delta R = R_0\alpha P$	P in N/m^2; $P>0$ stretch, $P<0$ squeeze
$P = \dfrac{F}{A}$	(pressure) = (force) ÷ (area)
$P = \dfrac{F\,L}{W\,H^2}$	cantilever: force F at the tip, length L, width W, thickness H
$V_{out} = \dfrac{V_0}{4}\alpha P = \dfrac{V_0}{4}\alpha\dfrac{FL}{WH^2}$	bridge output for a piezoresistor

Silicon: $\alpha \approx -1\times 10^{-9}$ m^2/N (usually negative).

Quick example	Work → answer
$V_0 = 4$ V reads 2 mV	$\frac{\Delta R}{R} = \frac{4\cdot 0.002}{4} = 0.002$
cantilever $F = 10^{-3}$ N, $L = 1$ mm, $W = 0.1$ mm, $H = 10\,\mu$m	$P = \frac{10^{-3}\cdot 10^{-3}}{10^{-4}\cdot 10^{-10}} = 10^8$ N/m^2
same, $V_0 = 1$ V, $\alpha = -10^{-9}$	$V_{out} = \frac{1}{4}(-10^{-9})(10^8) = -25$ mV

14. nonlinear elements: fuse, diode, LED [book 2-1.4, 2-6]

- **Fuse**: a wire that opens when the current passes its rating. A short at the load draws $I = V_s/R_s$, which blows it.
- **Ideal diode**: a one-way valve. Forward ($V_D > 0$): a short. Reverse ($V_D < 0$): an open, $I = 0$.
- **Real diode model**: off until V_D reaches V_F, then a drop of V_F (plus a small R_D, often ignored). Silicon: $V_F \approx 0.7$ V.
- **LED**: a diode that lights. On fully at about 20 mA. Red $V_F \approx 1.6$ V, green or blue ≈ 3.3 V. Curve: $I \approx aV_D^2$ ($V_D > 0$).

Formula	In words
$I = \dfrac{V_s - V_F}{R}$	forward diode in series with R
$R = \dfrac{V_s - V_F}{I}$	resistor to set an LED current
$R_D \approx \dfrac{\Delta V_D}{\Delta I}$	slope of the curve above V_F

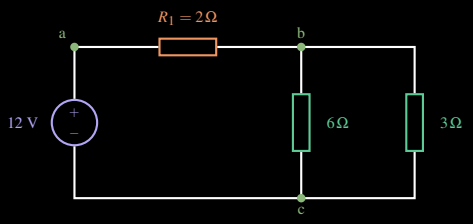
Diode procedure: (1) assume forward bias, (2) compute I, (3) if $I > 0$, done; if not, the diode is off and $I = 0$. For a square or triangle source, do each half of the cycle separately.

Quick example	Work → answer
5 V, 1 kΩ, Si diode forward	$I = \frac{5-0.7}{1000} = 4.3$ mA > 0 ✓
same source reversed (−5 V)	$\frac{-5-0.7}{1000} < 0 \Rightarrow$ off, $I = 0$
red LED at 20 mA from 5 V	$R = \frac{5-1.6}{0.02} = 170\,\Omega$

***I–V* slopes** [book Fig. 2-38]: resistor $1/R$, ideal voltage source vertical (∞), ideal current source flat (0).

15. putting it together: one full circuit

Reduce, find the source current, then expand back out with Ohm's law, KVL, and KCL.



Worked: find every current, voltage, and power

1. PIECES
6, 3 Ω parallel (share b, c) R_1 series 12 V

↓

2. COMBINE (§9)
$$\frac{6\cdot 3}{6+3} = 2\,\Omega; \quad R_{eq} = 2+2 = 4\,\Omega$$

↓

3. SOURCE CURRENT: $i = v_s/R_{eq}$
$$i = \frac{12}{4} = 3\text{ A}$$

↓

4. DROPS: $v = iR$**, THEN KVL**
$v_1 = 3\cdot 2 = 6$ V; $v_{bc} = 12-6 = 6$ V *(or $3\cdot 2$)*

↓

5. BRANCH CURRENTS: $i = v_{bc}/R$
$i_2 = \frac{6}{6} = 1$ A, $i_3 = \frac{6}{3} = 2$ A *(KCL: $1+2=3$ ✓)*

↓

6. POWER
$p_1 = 18$, $p_2 = 6$, $p_3 = 12$ W. Total 36 W $= 12\cdot 3$ from the source ✓